I.

1.

Reference solution: $y = C - \dfrac{2\pi^2 k}{mg}(x-a)^2$ （C is an arbitrary constant).

2.

Reference solution: $\dfrac{7}{12}R$.

3.

Reference solution: $30\ \mu\text{m}$.

4.

Reference solution:

$U_{\text{O}} = \dfrac{kT}{e}\ln\left(\dfrac{I_{\text{SC}}}{I_{\text{S}}}+1\right)$; e is electron charge magnitude, k Boltzmann's constant, and T is absolute temperature; I_{S} is the reverse saturation current of the p-n junction.

II.

Reference solution:

1. Scaling position and time as $\boldsymbol{r}' = \alpha \boldsymbol{r}$ 、 $t' = \beta t$ gives velocity and acceleration:

$$\boldsymbol{v}' = \lim_{\Delta t' \to 0} \frac{\Delta \boldsymbol{r}'}{\Delta t'} = \frac{\alpha}{\beta} \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{r}}{\Delta t} = \frac{\alpha}{\beta}\boldsymbol{v} \ , \tag{1}$$

$$\boldsymbol{a}' = \lim_{\Delta t' \to 0} \frac{\Delta \boldsymbol{v}'}{\Delta t'} = \frac{\alpha}{\beta^2} \lim_{\Delta t \to 0} \frac{\Delta \boldsymbol{v}}{\Delta t} = \frac{\alpha}{\beta^2}\boldsymbol{a} \ . \tag{2}$$

Both the original and scaled particle motion obey Newton's equations:

$$m\boldsymbol{a} = \boldsymbol{F}(\boldsymbol{r}) \ , \tag{3}$$

$$m\boldsymbol{a}' = \boldsymbol{F}(\boldsymbol{r}') \ . \tag{4}$$

Using (2) and $\boldsymbol{F}(\alpha \boldsymbol{r}) = \alpha^k \boldsymbol{F}(\boldsymbol{r})$, (4) becomes

$$m\boldsymbol{a} = \alpha^{k-1}\beta^2 \boldsymbol{F}(\boldsymbol{r}) \ . \tag{5}$$

Comparison with (3) gives

$$\beta = \alpha^{\frac{1-k}{2}} \ . \tag{6}$$

2. For gravity, $k = -2$. If the orbital size scales as

$$l' = \alpha l \tag{7}$$

its period scales as

$$\tau' = \beta\tau = \alpha^{3/2}\tau \tag{8}$$

and hence

$$\frac{\tau'^2}{l'^3} = \frac{\tau^2}{l^3}\,. \tag{9}$$

III.

Relative position is $\boldsymbol{r} = \boldsymbol{r}_\mathrm{B} - \boldsymbol{r}_\mathrm{A}$, with unit vector $\boldsymbol{e}_r = \boldsymbol{r}/r$. Particles are confined to their tubes; their net forces are the electrostatic components along their motion. Their equations of motion are

$$\begin{cases} m\boldsymbol{a}_\mathrm{A} = -\dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{i}\right)\boldsymbol{i} \\ m\boldsymbol{a}_\mathrm{B} = \dfrac{kq^2}{r^2}\left(\boldsymbol{e}_r \cdot \boldsymbol{j}\right)\boldsymbol{j} \end{cases}, \tag{1}$$

Subtraction gives

$$m\boldsymbol{a} = \frac{kq^2}{r^2}\boldsymbol{e}_r\,, \tag{2}$$

Here $\boldsymbol{a} = \boldsymbol{a}_\mathrm{B} - \boldsymbol{a}_\mathrm{A}$ is the acceleration of B relative to A. By (2), B relative to A moves as a particle of mass m in a fixed-centre Coulomb field, with potential energy

$$U = \frac{kq^2}{r}\,. \tag{3}$$

Angular momentum and energy are conserved in a central field. Both refer to B relative to A; the initial conditions give

$$L = \left|-md\boldsymbol{i} \times \left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)\right| = md\sqrt{\frac{kq^2}{md}}\,, \tag{4}$$

$$E = \frac{1}{2}m\left(-u_0\boldsymbol{j} + u_0\boldsymbol{i}\right)^2 + \frac{kq^2}{d} = \frac{2kq^2}{d}\,. \tag{5}$$

The closest distance is r_m; there the velocity u_m is perpendicular to $r_\mathrm{m}\boldsymbol{e}_r$, giving

$$mr_\mathrm{m}u_\mathrm{m} = md\sqrt{\frac{kq^2}{md}}\,, \tag{6}$$

$$\frac{1}{2}mu_\mathrm{m}^2 + \frac{kq^2}{r_\mathrm{m}} = \frac{2kq^2}{d}\,. \tag{7}$$

Solving the two equations gives

$$r_{\mathrm{m}} = \frac{1+\sqrt{5}}{4} d \ . \tag{8}$$

IV.

Reference solution:

For convenience, all heats below denote magnitudes.

Combine an engine and heat pump: the engine converts boiler heat to work, which drives the pump to absorb outdoor heat and deliver it indoors.

The boiler is the engine's hot source; a cold source can be indoors or outdoors. For the room, let the engine absorb Q_0 from the boiler and reject Q_{10} indoors. Then

$$\frac{Q_0}{T_0} - \frac{Q_{10}}{T_1} \le 0 \ . \tag{9}$$

The pump's hot source is indoors, its cold source outdoors. It absorbs Q_2 outdoors and rejects Q_{12} indoors. Then

$$-\frac{Q_{12}}{T_1} + \frac{Q_2}{T_2} \le 0 \ . \tag{10}$$

The room receives a total heat from the combined cycle of

$$Q_1 = Q_{10} + Q_{12} \ . \tag{11}$$

Adding (9)、(10) gives

$$\frac{Q_0}{T_0} + \frac{Q_2}{T_2} - \frac{Q_1}{T_1} \le 0 \ . \tag{12}$$

The same result follows if the engine uses outdoors as its cold source.

Energy conservation gives

$$Q_1 = Q_0 + Q_2 \ . \tag{13}$$

To deliver Q_1 directly to the room, the boiler must release

$$Q_0' = Q_1 \ . \tag{14}$$

Combining (12)、(13) gives the boiler heat for combined heating:

$$Q_0 \ge \frac{T_0\left(T_1 - T_2\right)}{T_1\left(T_0 - T_2\right)} Q_1 \ . \tag{15}$$

The fractional cost reduction is

$$\frac{Q_0' - Q_0}{Q_0'} \le \frac{T_2\left(T_0 - T_1\right)}{T_1\left(T_0 - T_2\right)} \ . \tag{16}$$

The theoretical limit is reached when equality holds.

V.

Reference solution:

Newton's equations for the two-dimensional electrons along x and y are

$$ma_x = -eE - ev_y B - \frac{mv_x}{\tau}\,, \tag{17}$$

$$ma_y = ev_x B - \frac{mv_y}{\tau}\,. \tag{18}$$

In steady state $\boldsymbol{v}$ is time-independent, so $a_x = a_y = 0$. Thus (18) gives

$$v_y = \omega\tau v_x\,, \tag{19}$$

Substitution into (17) gives

$$v_x = -\frac{e\tau E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{20}$$

By definition,

$$j_x = \frac{ne^2\tau E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{21}$$

$$j_y = \frac{ne^2\omega\tau^2 E}{m\left(1+\omega^2\tau^2\right)}\,, \tag{22}$$

Thus a current also flows perpendicular to the electric field.

VI.

Reference solution:

1. The electric field E between the plates is

$$E = E_0 \sin(2\pi z/\lambda_z)\sin(2\pi x/\lambda_x - \omega t)\,. \tag{1}$$

The plates are ideal conductors, so at z=0 and d, E=0. Hence

$$2\pi d/\lambda_z = m\pi, \quad \lambda_z = 2d/m, \quad m = 1,2,3\ldots\,. \tag{2}$$

2. Using the relation below, the wave in (1) can be

$$\sin\alpha\sin\beta = \frac{1}{2}\left[\cos(\alpha-\beta) - \cos(\alpha+\beta)\right], \tag{3}$$

expressed as the superposition of two plane electromagnetic waves:

$$\begin{cases} E = E_1 + E_2 \\ E_1 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x - 2\pi z / \lambda_z - \omega t) \\ E_2 = \dfrac{1}{2} E_0 \cos(2\pi x / \lambda_x + 2\pi z / \lambda_z - \omega t - \pi) \end{cases}, \tag{4}$$

Both waves have wavelength λ_0 given by

$$\left(\frac{1}{\lambda_0}\right)^2 = \left(\frac{1}{\lambda_x}\right)^2 + \left(\frac{1}{\lambda_z}\right)^2 = \left(\frac{\omega}{2\pi c}\right)^2 . \tag{5}$$

For m=1, $\lambda_z = 2d$,

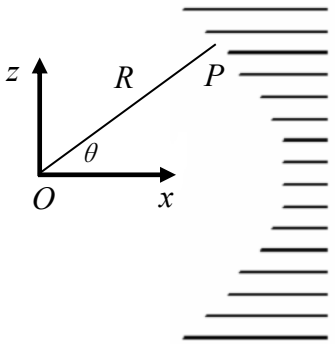
$$\lambda_x = \frac{1}{\sqrt{(2\pi c/\omega)^2 - (1/2d)^2}} . \tag{6}$$

3. Hence the wave speed along x, v_x , is

$$v_x = (\omega / 2\pi)\lambda_x = \frac{c}{\sqrt{1 - \left(\dfrac{\pi c}{\omega d}\right)^2}} . \tag{7}$$

Any P at the stack's left edge has source distance OP=R; OP is at θ to horizontal. A horizontally propagating plane-wave output requires

$$L / c = R / c + (L - R\cos\theta) / v_x , \tag{8}$$



This gives

$$R = L\left(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2}\right)\left(1 - \sqrt{1 - \left(\frac{\pi c}{\omega d}\right)^2}\cos\theta\right) . \tag{9}$$

VII.

Reference solution 1：

The following equations

$$F = ma\ , \tag{1}$$

$$a = \frac{v^2}{r}\ , \tag{2}$$

$$F = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r^2}\ , \tag{3}$$

give the electron's total energy:

$$U = \frac{-e^2}{8\pi\varepsilon_0 r_0}\ . \tag{4}$$

The condition $E_B = -U = 13.6\text{eV}$ gives hydrogen's orbital radius and speed:

$$r_0 = 5.29\times10^{-11}\text{m}\ , \tag{5}$$

$$v_0 = 2.19\times10^{6}\text{m/s}\ . \tag{6}$$

The Larmor formula gives the initial radiated power:

$$P = \frac{e^2a^2}{6\pi c^3\varepsilon_0} = \frac{e^6}{96\pi^3c^3\varepsilon_0^3m^2r_0^4}\ . \tag{7}$$

During a short interval Δt, radiation reduces the electron's total energy U by

$$\Delta U = -P\times\Delta t\ , \tag{8}$$

$$\Delta U = \frac{-e^2}{8\pi\varepsilon_0}\left(\frac{1}{r-\Delta r}-\frac{1}{r}\right) = \frac{-e^2}{8\pi\varepsilon_0 r^2}\Delta r\ , \tag{9}$$

Here Δr is the decrease in orbital radius. The relation between time and radius r is

$$\Delta t = \frac{12\pi^2c^3\varepsilon_0^2m^2r^2}{e^4}\Delta r = A\times4\pi r^2\Delta r\ , \tag{10}$$

where $A = \dfrac{3\pi c^3\varepsilon_0^2m^2}{e^4}$.

For a sphere of radius r_0, $4\pi r^2\Delta r$ is the volume of a thin shell at radius r. Summing from r_0 to 0 gives volume $\dfrac{4}{3}\pi r_0^3$. Correspondingly, the electron orbit shrinks from r_0 to 0 in a time

$$\tau = \sum \Delta t = \frac{4\pi^2 c^3 \varepsilon_0^2 m^2 r_0^3}{e^4} , \tag{11}$$

Substituting the data gives

$$\tau = 1.56 \times 10^{-11} \text{s} . \tag{12}$$

Reference solution 2：

For high-energy electrons,

$$v \approx c\,, \tag{1}$$

$$E = mc^2\,, \tag{2}$$

$$a = \frac{v^2}{R}\,, \tag{3}$$

$$F = ecB\,, \tag{4}$$

$$F = ma\,, \tag{5}$$

These conditions give the electron's bending radius:

$$R = \frac{E}{ecB}\,. \tag{6}$$

The number of electrons in the storage ring is

$$n = \frac{Q}{e} = \frac{I\Delta t}{e}\,, \tag{7}$$

where Δt is one revolution time. From (3) and the radiation condition, an electron loses a total energy per turn given by (there is no radiation on the straight sections)

$$E_{\text{turn}} = \frac{e^2 a^2 \gamma^4}{6\pi c^3 \varepsilon_0} \times \frac{2\pi R}{c} = \frac{e^2 \gamma^4}{3\varepsilon_0 R}\,. \tag{8}$$

Equations (7)、(8) give the total power dissipated by the ring electrons:

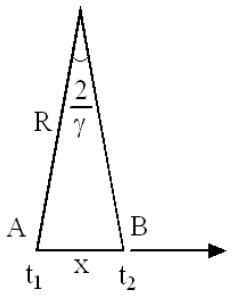
$$P_{\text{total}} = \frac{nE_{\text{turn}}}{\Delta t} = \frac{Ie\gamma^4}{3\varepsilon_0 R} = 4.34\times 10^5\,\text{W}\,. \tag{9}$$

The power at each outlet is

$$P_{\text{O}} = \frac{P_{\text{total}}}{N} = 2.17\times 10^4\,\text{W}\,. \tag{10}$$

Reference solution 3：

At *P* along the tangent to an electron orbit, an observer sees radiation only briefly. This is the arrival-time interval of light emitted as the velocity direction sweeps the cone angle $2/\gamma$. The electron travels the following distance during that interval:



$$x = \frac{2R}{\gamma} . \qquad (1)$$

Let $\beta = v / c$. The interval between emission at the two points is

$$t_2 - t_1 = \frac{2R}{\gamma c \beta} . \qquad (2)$$

For the tangent observer, subtract the travel time of light from *A* to *B* from this interval, as given below:

$$t' = \frac{2R}{\gamma c} , \qquad (3)$$

Then

$$\Delta T = t_2 - t_1 - t' . \qquad (4)$$

Also,

$$\gamma = \frac{mc^2}{m_0 c^2} = \frac{1}{\sqrt{1-\beta^2}} , \qquad (5)$$

In the limit $1-\beta \to 0$,

$$\frac{2}{\beta}(1-\beta) = \frac{1}{\gamma^2} . \qquad (6)$$

Substitution into (4) gives

$$\Delta T = \frac{R}{c\gamma^3} . \qquad (7)$$

The numerical data give

$$\Delta T = 9.51 \times 10^{-20} \text{s} \qquad (8)$$